

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

Academic Year: 2025-26

Year: Second Year

Semester: II

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Subject: Python Programming

Assignment No.: 25

Date: 08/04/2026

Lab Assignment: 25

Aim:

To write a program that accepts a filename from the user, opens the file, and counts the number of times a character appears in the file.

Objective:

- To understand file handling with user input in Python.
- To learn how to read data from a file.
- To count the occurrence of a specific character in a file.
- To improve skills in string and file processing.

Theory:

File handling in Python allows users to perform operations such as reading and writing data from files. Using the `open()` function, a file can be accessed in different modes like read ('r'), write ('w'), etc. In this program, the user provides the filename, and the file is opened in read mode. The content of the file is then read using methods like `read()`. Once the content is obtained, it can be treated as a string. To count the occurrence of a specific character, Python provides the `count()` method, which returns the number of times a character appears in a string. This makes it easy to analyze file content efficiently. Proper handling also includes closing the file after operations using the `close()` method to free system resources.

Algorithm:

1. Start the program.
2. Accept the filename from the user.
3. Open the file in read mode.
4. Read the content of the file.
5. Accept a character from the user.
6. Count the number of occurrences of the character using `count()`.
7. Display the count.
8. Close the file.
9. End the program.

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Pseudo Code:

```
START
INPUT filename
OPEN file in read mode
READ file content
INPUT character to count
COUNT occurrences of character
PRINT count
CLOSE file
END
```

Code:

```
# Take filename from user
filename = input("Enter file name: ")

# Take character to search
char = input("Enter character to count: ")

try:
    # Open file
    file = open(filename, "r")

    # Read content
    content = file.read()

    # Count occurrences
    count = content.count(char)

    print("Character", char, "appears", count, "times")

    # Close file
    file.close()

except FileNotFoundError:
    print("File not found")
```

Output:

```
Enter file name: Sample.txt
Enter character to count: l
Character l appears 3 times
```

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Conclusion:

Thus, the Python program was successfully executed to accept a filename from the user, open the file, and count the number of times a specific character appears in the file. This demonstrated the use of file handling and string operations for data analysis.